DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
West Point, New York 10996

MADN-CBE 22 May 2026

MEMORANDUM FOR RECORD

SUBJECT: Chemical Engineering Program Assessment and AAR (AY26-2)

1. The junior and senior faculty of the chemical engineering program at the United States Military Academy met on 22 May 2026 to conduct a post-semester review of courses taught during Spring of 2026 and a program assessment. The faculty members in attendance were Prof. Andrew Biaglow, COL Corey James, LTC (R) Sam Cowart (virtually), Dr. Enoch Nagelli, Dr. Simuck Yuk, MAJ Samuel Lowell, MAJ Louis Tobergte, MAJ Nijel Rogers, MAJ Josh Frey, MAJ Liz Golonski, and CPT(P) Chris Stewart, and CPT Denis Gilinski. LTC(R) Cowart attended virtually since he retired and departed USMA in March 2026.

2. Each course director presented topics and relevant assessments to their specific courses. These topics are outlined in more detail in the individual course assessments. These end of semester discussions serve as a tool to gauge effectiveness of course content and administration in meeting the ABET student outcomes specific to each course. Comments and questions from these discussions support the more formal content of the written course assessment packages that are completed at the end of each academic term. Ideas for course improvement and potential future changes are the focus of the discussions.

3. The following courses were discussed: CH300 Biomedical Engineering, CH362 Mass & Energy Balances, CH364 Chemical Reaction Engineering, CH367 Intro/Automatic Process Controls, CH400 Chemical Engineering Professional Practice, and CH402 Chemical Engineering Process Design were reviewed and thoroughly discussed by all attendees. Some of the topics led to broader discussions relevant to the entire program. These included: 1) Curriculum integration by defining the role and application of CHEMCAD to provide connection of foundational theory to chemical engineering applications for cadets, 2) The decline in process communication through P&IDs and the need to having dedicated portions of lessons in CH367, 3) The need for oral proficiency exams or evaluations to gauge cadets learning in specific course graded events, 4) The integration of dynamic controls with time based control models instead of Laplace, and 4) The decline in overall technical communication in writing events for cadets likely due to reliance on AI tools.

4. The point of contact for this document is the undersigned at enoch.nagelli@westpoint.edu or 845-938-3904.

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Chemical Engineering

Course and Program AAR (AY26-2)

Dr. Enoch A. Nagelli and Dr. Simuck F. Yuk

21 May 2026

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Agenda

- CH367
- CH362
- CH364
- CH400
- CH402
- CH300
- CH450
- Chemical Engineering Program AAR Comments

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CH367 AAR

Sustain:

- ✓ Capstone with CH364.
- ✓ Emphasis on fundamentals, like modeling, correlating processes to block diagrams, and P&IDs.
- ✓ Daily quizzes. Consider doing these at the beginning of class, no resources allowed. With the integration of AI into all that we do, the Cadets are using it to answer questions on the quizzes. We could better test their comprehension of the reading by doing it in class and it could still be done on Canvas.
- ✓ Continue use of SSI for describing/demonstrating dynamic behavior, stability, controller design, and tuning. The Cadets get a lot out of SSI and it allows you to start relatively simple and scale up.

Improve:

- ✓ One problem should be added to the TEE to increase breadth slightly; recommend a tuning/stability problem.
- ✓ Introduce python and ChemCAD to model controllers and control loops. This was accomplished, but still needs work. Start the process of upgrading the textbook to support this.
- ✓ Improve the controls portion of the Capstone. Improve format/quality of block diagrams and P&IDs, and in general provide more guidance.
- ✓ Consider integrating SSI into a few homework problems; guided simulation and analysis.

CH362 AAR

Course average: 83.8% (*AY25-2: 83%); * TEE average 75% (*AY25-2: 70%*)

Sustains

- **Maintain majority of points from in class assignments (74% this year)**
  - **Largely maintain overall structure – but make time for cadets to brief the PS solutions and/or add short in-class concept quizzes on Canvas**
- **Maintain hands-on lab and optimization labs**
- **Performance on TEE 75% increased (71% in AY25; 1's & 0's) - bonus point oral briefs of PS 8 and WPR 3 corrections, and a TEE review lesson all assisted cadets in tying concepts together**
- **Provide old graded events for cadets to use to practice; cadets studied much more using these old graded events and it levelled the playing field**

Improves

- **Split Capstone into individual writing assignment and group modelling project**
  - **Individual writing assignment: cadets dive deeply into a topic that interests them**
  - **Group modelling project: through a video tour or trip section, whole class learns about a ChemE process – they model it in groups and then experiment in ChemCAD to optimize the system – this would introduce them to design and optimization**
- **Add more oral briefs for points – some cadets could do the math, but didn't understand the concepts, and vice-versa; 3 trials of oral briefs this year all yielded substantial development**
- **Return WPRs to 55-minute lessons – 120-minute WPRs did not lead to better results or understanding**
  - **Use additional lab time to experiment in ChemCAD, as cadets never grasped the power of the software or how a ChemE would use it**
- **Poor Performance on WPR3 67% (88% in AY25; 1's & 0's) - cadets struggled with tying M&E balances together in a short time period**
- **Emphasize self-study using learncheme.com**
- **Consider developing a new TEE – current TEE emphasizes PV work 4 times**

CH364 AAR

Section III AY26-2 Course Assessment

- Course average: 88.4% (*AY25-2: 86.7%);* TEE average 82.5% (*AY25-2: 81.9%*)

Sustains

- Combined capstone project for both CH364/CH367.
- Dedicated Lessons and Lab periods as capstone working sessions.
  - Capstone IPR performance improved to 88% (1’s & 0’s)
- Performance on WPR2 (88%) and WPR3 (84.5%) increased (1’s & 0’s).
- Improved performance on Selectivity with multiple rxns Problem C on TEE 93% (83% in AY25; 1’s & 0’s)

Improves

- Average on Problem A FEE Style Questions on TEE declined 69% (AY25-2: 72.2%; from 1’s & 0’s).
  - Add Multiple choice FEE questions to WPRs
- Tie in BH sub-basement Bath and CSTR reactor experiments to labs 3 & 4.
- Poor performance on P&IDs on capstone.
  - Use Capstone Working Periods to model P&ID using LucidChart on Canvas
- Introduce cost estimation in CHEMCAD for reactor sizing with each lab.
- Develop additional PS problems, with increased reliance on numerical solvers and multiple concepts.

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CH400 AAR

Sustain:

- 19/25 Cadets Passed the FEE Overall. 76.0% Pass Rate
- No 2nd Attempts
  - 6 Failures Total: Faherty, Leja, Ebiuwhe, Macune, Stokes, Tringali
- 2 lessons per topic: CRE, Thermo, Controls, Transport, Chem/Bio, Seps
  - 1st lesson is practice problems & FE Ref Manual, 2nd lesson is Quiz
- Used Canvas Quizzes as timed quizzes (25 min) with FEE references
  - Cadets came to class and took a “topic FEE” with immediate results
  - Instructor and cadets received immediate feedback on performance
  - Instructor worked specific problems with remainder of class time
- Continue SSI modules: DIST & ACU
- 2x practice WPRs and full WPR; good recap of all topics in FEE format.

Improve:

- Trim question banks in PPI modules for more focused problem sets
- Re-write the Ethics quiz; questions nuanced / not directly from FEE manual
- Add additional SSI-Simulator Exercises
  - ACU Exercise with scenarios and/or CTOW with Level Controller

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CH402 AAR Section III (AY26-2)

<u>Sustain</u>:

Capstone project - Use AIChE design problem if possible; recent problems are good fit with curriculum, but not this year. If possible, use multiple problems.

Time survey quizzes, design groups based on grades, and field trip were good ideas.

<u>Improve</u>:

Control – Knowledge of control schemes and ability to draw P&IDs could be better. Some groups started early but some did not make sense (49% safety and 59% controls compared to 73% and 70% last AY, from 1/0's)

Reports – Cadets need a report template. LyX taught in CH365.

Research – Special topics research was poor 57% vs 95% last AY.

<u>Research</u>:

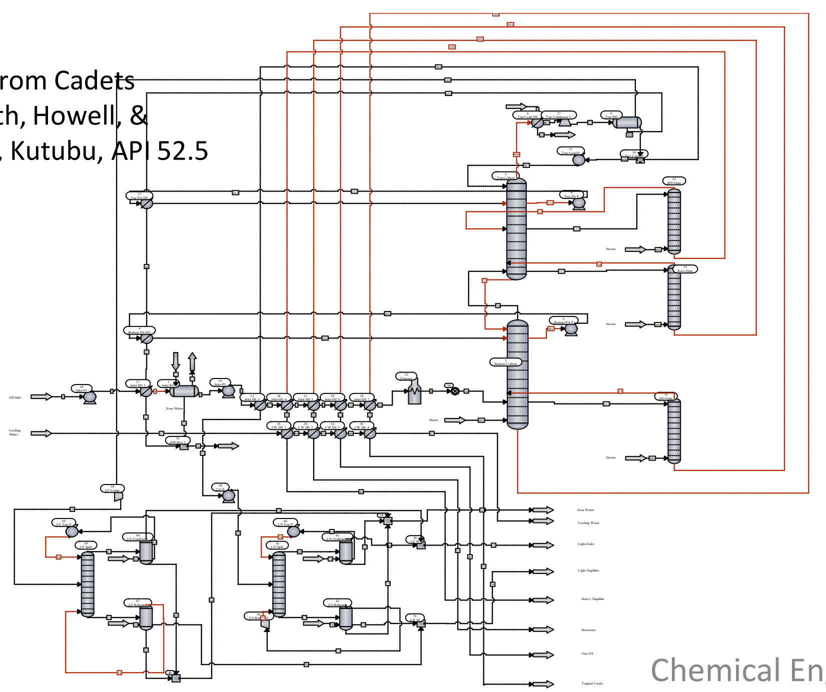
Discussion of boiling point analysis for all groups was not strong. This is a big improvement from last year.

Research on sulfur and water content to define design specs was good.

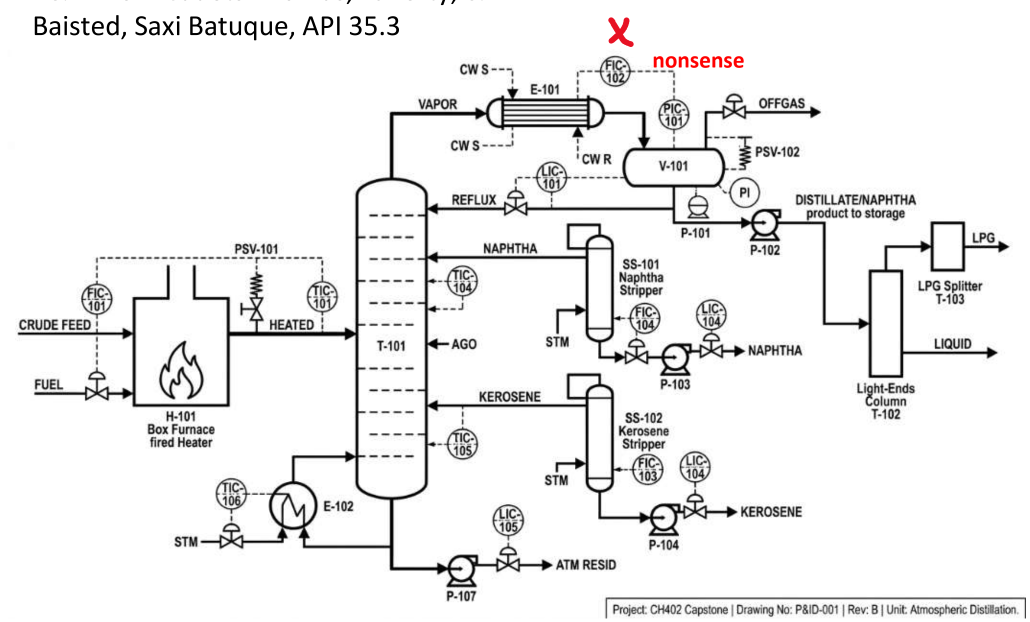
Research on context and social issues was good.

Research on special topics was generally poor. Suggest changing the topic by giving them a pic list of crude oils and asking them to design a process PLUS a feed cocktail that enables them to meet BP specs for JP8 instead of kerosene.

CC PFD from Cadets Mantooth, Howell, & Ferrying, Kutubu, API 52.5



P&ID from Cadets Thomas, Faherty, & Baisted, Saxi Batuque, API 35.3



CH300 AAR

- Course average: 91.9%; TEE average 88.8%

Sustain:

- ✓ Continue using in-class activities to guide students through formulating governing equations and deriving quantitative results.
- ✓ Capstone project on the bio scaffold has been shown as the most favorite graded component. Cadets displayed good engagement, particularly when exploring practical biomedical applications (AY25-2: 94.7%; AY26-2: 94.6%).

Improve:

- ✓ Align in-class coding exercises with contemporary subject matter to keep the material relevant (i.e., mRNA vaccine distribution/automated insulin pumps can be related to ODEs).
- ✓ Broaden the problem-solving exercises to emphasize numerical solvers and cross-concept integration.
- ✓ Improve the application problems in the Term-End Exam (TEE) to better assess cadets' ability to apply bioengineering fundamentals in practical contexts (AY25-2: 87.2%; AY26-2: 88.8%).

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CH450 AAR

- Course average: 92.1%

Sustain:

- ✓ Continue incorporating in-class activities to illustrate the formulation of governing equations and the process of obtaining quantitative results since the class population is diverse (AY26-2: 10 CEN, 7 BIO, 1 EVE).
- ✓ Maintain the use of end-of-block quizzes to truly assess cadets' comprehension of key learning objectives:
  - AY25-2: Quiz 1: 100.0%, Quiz 2: 95.2%, Quiz 3: 97.6%, Quiz 4/5: 100.0%
  - AY26-2: Quiz 1: 92.6%, Quiz 2: 89.4%, Quiz 3: 89.0%, Quiz 4: 91.1%, Quiz 5: 92.2%

Improve:

- ✓ Expand the in-class coding exercises with the contemporary subject matters (i.e., in-depth focus on pharmokinetic models or biosignaling).
- ✓ With the advent of generative A.I., having the oral examination as part of either end-of-semester Captone Project or TEE would be beneficial to assess the cadets' fundamental understanding of bioengineering principles.

Chem E Prog AAR

1. ABET e-Coursebooks/Documents will be priority in Summer
   - On-Site Visit on 30AUG-02SEP26
   - All Course Assessments Scrub (Dr. Nagelli & Dr. Yuk)
     - CH363 & CH383 from AY26-1
   - AY26-2 CD's work on e-Coursebooks
   - Faculty Meetings with Program Evaluator (31AUG)

2. ABET Lab Safety (POC: Dr. Yuk with with LTC Jerke, Safety OIC)
   - SB51/53 Lab (Clean up, SDS, Fire extinguisher, Fume hood)
   - CH101, CH383, CH275 Labs

3. Course Directors Handbook for all administrative guidance and SOP
   - EXSUM on Dept Sharepoint to include Course AAR Slides
     - Course Folder on Sharepoint (Rosters – alpha & OML, Course Assessment, all Course Materials (SIS, lesson schedule/objectives, grades etc.)
     - Graded Events Folder on Sharepoint (all graded events, solutions, and cut scales (labs, quizzes, WPRs, TEE, capstone, etc.)
   - Course Assessment Due 30 days after semester completion

Chem E Prog AAR

4. CH459 Summer Training (Dr. Yuk + Junior Faculty Members?)
    - Steam generator for EVAP
    - Running all unit operations
    - Lab safety and cleaning for ABET Walkthrough

5. CH485 Summer FDW (LTC Plante + Dr. Nagelli)
    - LTC Plante is CD for CH485 in AY27-1
    - Run all labs for CH485 and practice lessons with senior faculty in attendance (Prof. Biaglow, Dr Nagelli, and Dr. Yuk)

6. Conferences
    - AICHE Fall 2026 in Minneapolis, MN (NOV26) – Undergrad poster competition for cadets to compete (1 poster per cadet; S: SEP 2026) (POC: CPT Glinski as OIC)
        - Faculty & Grad Oral Talk Abstracts Submitted: 1LT Eddie Chen, Dr. Yuk, & Dr. Johnson

7. Program Led "events"
    - Chili Cookoff (OCT/NOV26, OIC: MAJ Frey, AOICs: MAJ Rogers, CPTs Stewart & Glinski)

8. Calendar Events
    - 29MAY: Dept On-Site (Conf Rm; APs, Assoc Prof and higher)
      Topics:
        - *Current Major Requirements*
        - *Minimum Major Requirements*
        - *ABET Assessment Requirements*
        - *Lab Consolidation Plan/Tax?*
        - *Department Events: AAR/Reassess*
    - 21JUN: Incoming Faculty Report Date
    - 22-26JUN: New Faculty In-processing
    - 29JUN: Class of 2030 R-Day
    - 29JUN-24JUL: FDW
    - 6-27JUL: CH101 STAP3

ABET Self-Study & On-Site Visit Timeline
Visit Dates: 30AUG-02SEP26

Current Accreditation

ABET Engineering Accreditation Commission	Accredited to September 30, 2027
ABET Computing Accreditation Commission	Accredited to September 30, 2027
ABET Applied and Natural Science Accreditation Commission	Accredited to September 30, 2032

Next General Review

Accredit to September 30, 2027. A request to ABET by January 31, 2026 will be required to initiate a reaccreditation evaluation visit. In preparation for the visit, a Self-Study Report must be submitted to ABET by July 1, 2026. The reaccreditation evaluation will be a comprehensive general review.

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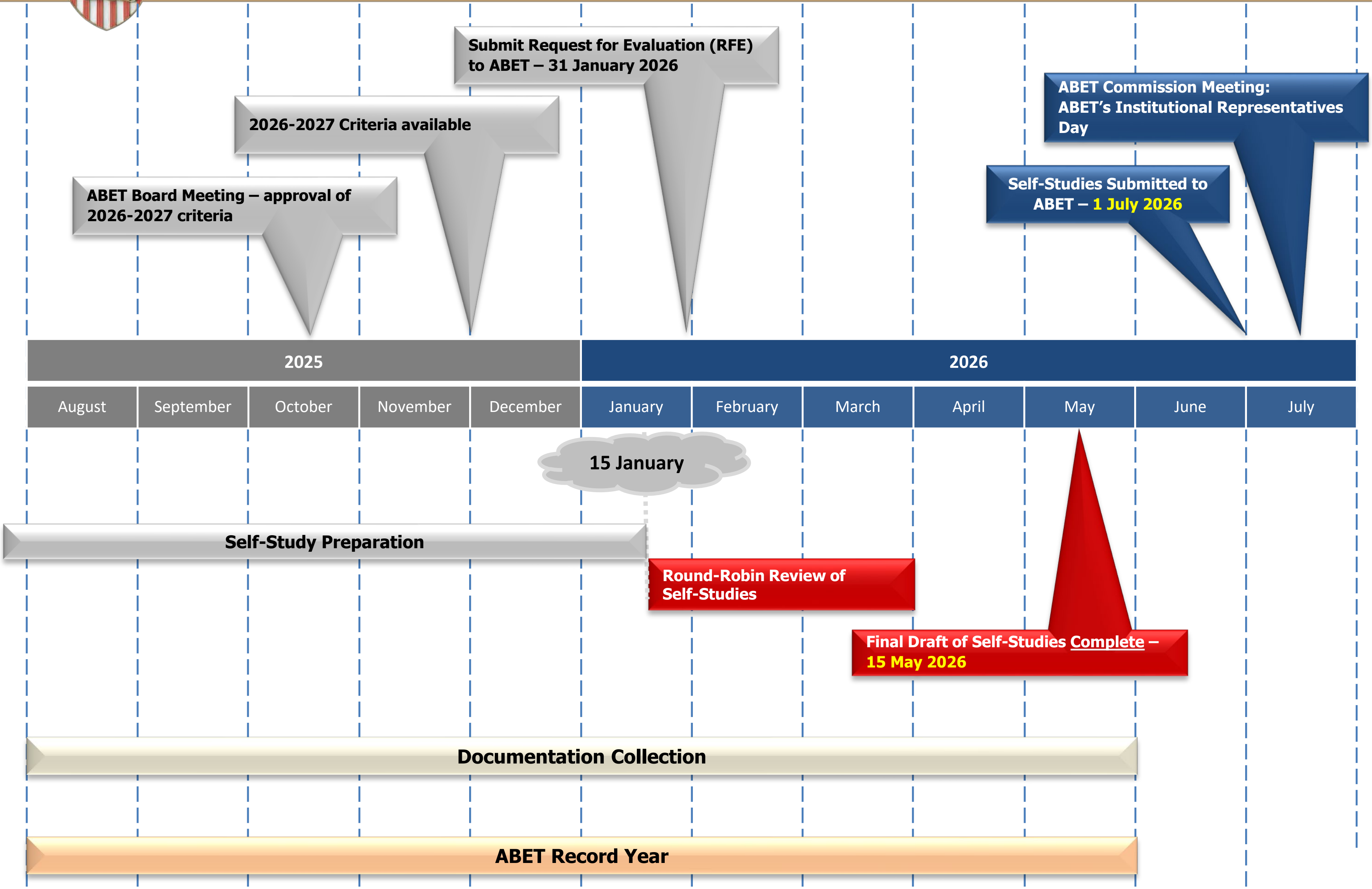
Record Year Timeline



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To-Do's

- Approve TC (USMA Level)
- Approve PEVs when assigned (USMA Level)
- Finalize Self-Studies – check for completeness and prepare for submission
- Submit Self-Studies
- Finalize all ET credit review memos with AARs and ensure updated Redbook Entries
- Finalize and complete external access site for display materials
- Complete adjustments to web pages
- Brief new senior leadership on ABET and ensure visit is scheduled on calendars
- Verify fire extinguishers and any other lab safety is updated
- Receive common e-Coursebooks for display from core courses
- Complete Logistics
  - Schedules for Team, TC, PEVs
  - Ground Transportation on-post
  - Tour of West Point (Jenn Voigtschild)
  - Legal Review for Tues lunch
  - Planning for Lunch at Club Monday
  - Complete purchase process to pay for the visit
- Positive communication with team prior to visit

Individual Program Prep

- Complete Self-Studies
- Complete record-year assessments and data collection
- Organize all policy memos etc.
- Organize your display materials and be prepared to put onto site for outside access
- Positive communication with PEVs (primary and alternates throughout the summer)
- USMA will provide transcripts on request
- Prepare your Faculty (all faculty and leadership)
- Prepare your lab staff
- Prepare your students
- Reach out to Advisory Board Members who may be able to come on Monday Lunch